

Temporal Video Grounding — Object Category

Approach, experiments, and reasoning behind the design decisions

Pipeline: `object_pipeline.py` (SigLIP + Grounding DINO frame scoring) → `build_submission.py` (score-to-interval postprocessing) | 24 object queries, 8 surveillance videos

1. Task and category scope

The task is temporal grounding: given a natural-language query and a video, predict the time interval(s), if any, where the query holds. Scoring is temporal-IoU F1 averaged over thresholds 0.3/0.5/0.7, macro-averaged over three query types (object, state, action). This report covers the **object** category only (24 of 72 queries) — the category attempted first because it reduces to a per-frame appearance question, unlike state (compound scene predicates) or action (requires temporal/motion reasoning).

2. Approach

2.1 Model choice

SigLIP (google/siglip-base-patch16-224) was chosen over CLIP because its sigmoid loss produces a calibrated image-text similarity rather than a purely rank-based one, making an absolute cutoff (“above X counts as present”) more meaningful than with plain CLIP cosine similarity. **Grounding DINO** (IDEA-Research/grounding-dino-tiny) is a zero-shot phrase-grounding detector: given an image and a batch of query phrases for that video, it returns candidate boxes with a confidence score, letting us localize *where* in the frame the query might be, not just whether the whole frame resembles it.

2.2 Why not rely on the whole frame alone

Object queries typically name a small part of the frame (a person, a backpack, a single car among many). A single whole-frame embedding is dominated by background, so the pipeline additionally crops each detected box (with a 10% margin) and re-embeds *only that crop* with SigLIP, comparing it again to the query text. Three signals are produced per sampled frame (every 0.5s): `global_score` (whole frame), `detector_score` (Grounding DINO confidence), `crop_score` (SigLIP on the cropped box). Section 3.5 revisits and removes the first of these.

2.3 Fusion

The three signals live on incomparable scales (detector confidence is roughly 0–1; SigLIP cosine similarity sits in a narrow 0–0.2 band). Each is **z-score normalized per query, across that query's own timeline** ($z = (x - \text{mean}) / \text{std}$) before being combined in a weighted sum. This is necessary to fuse the signals at all, but it has a structural side effect that drives most of Section 3: a z-score is always relative to that query's own noise floor, so it will report a “peak” even when the underlying signal never contains real evidence.

3. Experiments and design iterations

3.1 Fixed threshold → over-fragmentation → hysteresis

The first version thresholded the smoothed fused score at a single fixed z-value. For queries where the object is visible almost continuously, frame-to-frame detector noise made the score oscillate across that single cutoff dozens of times, producing up to 15 tiny predicted intervals for one real, continuous appearance — costly under IoU-matched F1, where every extra interval is an unmatched false positive. Fix: replace the single threshold with **hysteresis** — a frame only *enters* an event above an ENTER bar (0.75) and only *leaves* it below a lower EXIT bar (0.15), so noise straddling one boundary can no longer flip the state back and forth. Combined with a rolling-mean smoothing pass and merging of same-event segments separated by short gaps, this took queries like “a blue umbrella” from 15 fragments down to 1–4 plausible ones.

3.2 The pipeline could not say “absent”

The evaluation formula explicitly scores empty-ground-truth queries (a query whose object never appears, correctly predicted as NONE, scores 1.0). But per-query z-scoring means *any* curve has a relative peak, even pure noise. Three queries confirmed this directly: q011, q030, q046 never produced a single Grounding DINO detection across their entire video (detector_score = 0 for thousands of sampled frames), yet the fused z-score still produced confident-looking peaks (1.07, 1.18, 2.04). Fix: an **absolute evidence gate** on top of the relative hysteresis segments — a candidate interval is only kept if it contains at least one frame with a real detection (detector_score > 0, which itself already means Grounding DINO’s own confidence threshold was cleared) *and* a crop SigLIP similarity of at least 0.085 (confident true matches in this dataset cluster around 0.11–0.17; unsupported ones sit at 0.05–0.09).

3.3 Manual verification against real frames

Numeric thresholds were sanity-checked by extracting the actual video frame at each candidate peak. This caught a real design bug and one dataset-level limitation.

Query	Claim checked	Frame evidence	Outcome
q002	“a cyclist”	Cyclist clearly visible at the scored peak time.	Confirmed true positive.
q003	“a yellow car”	Peak box is a beige/tan sedan, not yellow.	Likely false positive — color-attribute confusion.
q021	“a blue umbrella”	Every umbrella in the scene is red/terracotta.	Likely false positive — same failure mode.

Frames referenced above are saved under `outputs/frame_checks/` (`q002_cyclist_top.png`, `q003_exact_box.png`, `q021_box_86.png`).



Figure 1. q002 top-scoring frame (video_01, t=114.0s) — a cyclist is plainly visible on the sidewalk, confirming the detector+crop signal is grounded, not noise.



Figure 2. Left: the exact detector box scored highest for q003 (“a yellow car”) is a beige sedan (crop_score=0.076, notably below the confident-match range). Right: the exact box for q021 (“a blue umbrella”) is a red umbrella (crop_score=0.091). Both sit right at the evidence-gate boundary, which cosine similarity alone does not resolve cleanly.

3.4 Attempted fix: pixel-level color verification (reverted)

Since SigLIP cosine similarity does not cleanly separate “right object, wrong color” from a genuine match (both land around 0.08–0.10), the next idea was to verify color directly from pixels: read the median HSV hue/saturation/value inside the detected box and classify it into a color bucket (red/yellow/blue/...), then require it to match a color word extracted from the query text. This was implemented and immediately reverted after it rejected q001 (“a man with a red backpack”) — a query manually confirmed present in the video. Debugging showed why: the highest-crop-score box for q001 was 18×22 pixels and captured hair/shadow, not the backpack, giving a misleading median hue; separately, a box for “a yellow dumpster filled with sand” captured mostly the sand rather than the dumpster’s own color. A single whole-box median color is too easily dominated by background, occlusion, or the wrong sub-region to gate a hard pass/fail on. This is left as a limitation rather than solved.

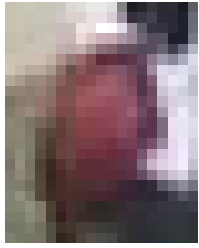


Figure 3. Left: the entire 18×22px box used to judge q001's color — mostly hair, not backpack. Right: the box for the “yellow dumpster” query, mostly filled with sand rather than the dumpster body. Both make a single median-pixel color check unreliable.

3.5 Ablation: removing the global whole-frame signal

A further look at the fusion design raised a structural question: the whole-frame (“global”) SigLIP signal was originally included as a fallback for frames the detector misses. But for a genuinely small object, the whole-frame embedding is diluted by background for the *same underlying reason* the detector misses it — so it is not an independent fallback, and could inject noise exactly where nothing else provides evidence.

This was tested empirically rather than assumed: the z-score fusion was recomputed from the already-cached raw detector_score/crop_score columns, once with the original weights (global 0.20 / detector 0.30 / crop 0.50) and once with global dropped (detector 0.375 / crop 0.625, preserving their 3:5 ratio) — no re-inference required.

Table 1. Queries with zero Grounding DINO detections across their entire video.

Query	Text	Max detector	Max fused, with global	Max fused, without global
q011	a person wearing a yellow shirt	0.00	1.066	0.000
q030	a man wearing a blue jacket and purple pants	0.00	1.184	0.000
q046	a person wearing a red cap	0.00	2.040	0.000

Without the global term, the fused score for all three is a flat 0.000 — there is nothing else left to produce a peak from. With it, every one of those “peaks” (1.07, 1.18, 2.04) was manufactured entirely by whole-frame embedding drift, unrelated to the query.

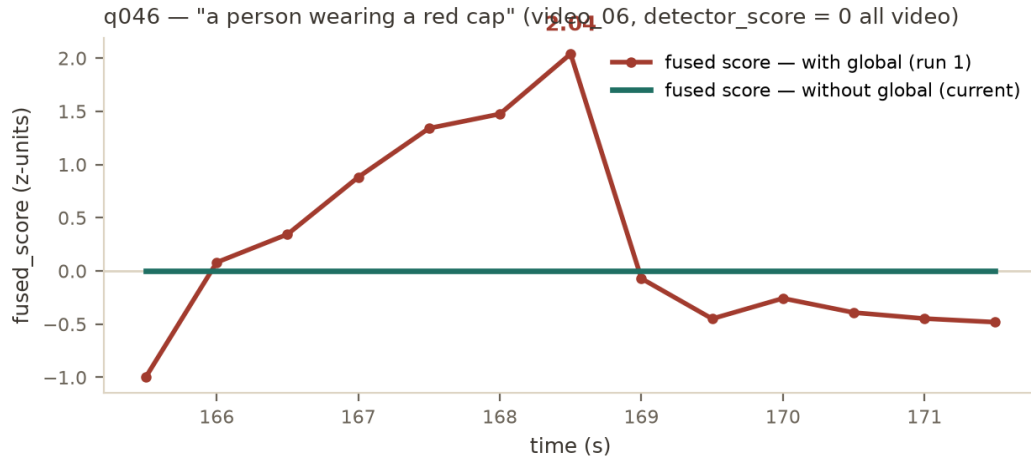


Figure 4. Real fused_score values for q046 in a representative 6-second window. The “with global” curve is what run 1 actually produced (climbing to 2.04 with no detection nearby); “without global” is flat because detector_score is a constant zero for this query for the full video.

Across all 24 object queries, the two fusion variants were compared by Pearson correlation of their time series. Most queries correlate above 0.9 (detector+crop already carry the signal; global changes little either way); the three zero-detection queries above cannot even be correlated, since one side is a constant. Re-running the full postprocessing pipeline (hysteresis + evidence gate) on both variants produced the same final kept/NONE verdict for all 24 queries — the evidence gate from Section 3.2 already independently protects against this failure mode. The practical effect of removing global was therefore mainly a loss of incidental curve-smoothing for a few already-detected queries (more fragmented candidate segments), which was compensated by widening the smoothing window (5→7 samples) and the segment-merge gap (3.0s→5.0s) in build_submission.py, rather than by keeping a signal whose justification did not hold up.

4. Final parameters (object_pipeline.py / build_submission.py)

Parameter	Value	Role
Fusion weights	detector 0.375 / crop 0.625	No global term (Section 3.5).
Sample interval	0.5 s	Frame sampling rate for both SigLIP and Grounding DINO passes.
Grounding DINO thresholds	box 0.25 / text 0.20	Detector’s own confidence cutoffs.
Crop margin	10%	Padding around a detected box before re-cropping for SigLIP.
Smoothing window	7 samples (~3.5s)	Rolling mean on fused_score before segmentation.
Hysteresis ENTER / EXIT	0.75 / 0.15 (z-units)	Prevents threshold-flapping from noise (Section 3.1).
Max merge gap	5.0 s	Bridges short dropouts inside one continuous event.
Min interval duration	1.0 s	Drops single-sample noise spikes.

Evidence gate	<code>detector_score > 0</code> and <code>crop_score ≥ 0.085</code>	Requires real detection, not just a relative peak (Section 3.2).
Boundary padding	<code>±0.25 s</code>	Half a sample step, since the true edge lies between samples.

5. Results and known limitations

16 of 24 object queries clear the evidence gate and receive one or more predicted intervals; 8 (q003, q011, q020, q030, q046, q047, q048, q056) are predicted NONE. For q011/q030/q046 this is high-confidence (zero detections all video). For q003 (color mismatch, Section 3.3), q020/q047/q048/q056 (weak but nonzero crop evidence, 0.052–0.077) it is a lower-confidence call the pipeline cannot fully verify without ground truth — these may be real true negatives, or real objects our detector/crop combination is simply too weak to confirm confidently.

The main known limitation is **color-attribute binding**: SigLIP/CLIP-style cosine similarity does not reliably distinguish “right object, wrong color” from a genuine match when the color is visually adjacent (beige vs. yellow, red vs. blue in a red-dominant scene). A pixel-level fix was attempted and reverted (Section 3.4) after it proved less reliable than the embedding itself on small or mixed-content boxes; this remains open for the object category, where almost every query is distinguished primarily by color.

State and action queries (48 of 72) are not yet implemented in this pipeline and are currently written as NONE placeholders in submission.csv so the file matches the required format.